REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Fashion Design & Apparel Manufacture Lab

A full-screen, syllabus-style digital handbook for Course Code 6067. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6067

COURSE TITLE
Fashion Design & Apparel Manufacture Lab

DEPARTMENT
Textile Technology

TYPE
Lab

SEMESTER
6

CATEGORY
Practical

USE
Study + notes PDF

FORMAT
Big handbook

6067

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**	**Module lessons**	**4 detailed units**
Formula bank	**calculations**	**Practice bank**	**questions + tasks**
Model paper	**official style**	**Answer key**	**full points**

Course objectives and syllabus map

Objectives

- ✓ Practise fashion sketching, measurement and pattern drafting
- ✓ Construct basic garments using correct cutting and sewing method
- ✓ Prepare finishing, inspection and costing records

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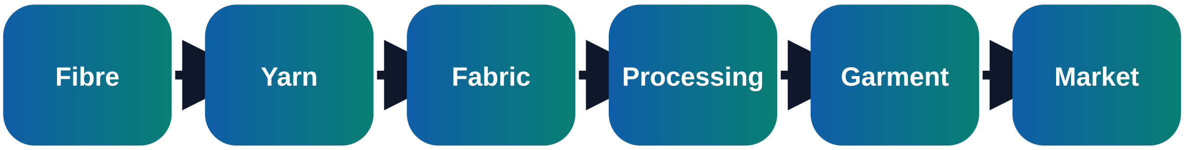
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Fashion design preparation	Elements of design: line, shape, colour, texture and balance; Fashion figure, garment sketching and design board	Short notes, calculations, diagram, checklist, viva answers
M2	Measurement and pattern drafting	Body measurement points and allowance; Basic bodice, sleeve, skirt and trouser block principles	Short notes, calculations, diagram, checklist, viva answers
M3	Cutting and sewing operations	Fabric spreading, marker planning, cutting and bundling; Machine selection, stitch class, seam class and needle selection	Short notes, calculations, diagram, checklist, viva answers
M4	Finishing, inspection and costing	Trimming, pressing, folding, labelling and packing; Measurement checking and visual inspection	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Fashion design preparation

Core explanation

- ✓ Elements of design: line, shape, colour, texture and balance
- ✓ Fashion figure, garment sketching and design board
- ✓ Fabric selection based on drape, end use and style
- ✓ Mood board, colour palette and design specification sheet

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Measurement and pattern drafting

Core explanation

- ✓ Body measurement points and allowance
- ✓ Basic bodice, sleeve, skirt and trouser block principles
- ✓ Pattern layout, grain line, seam allowance and notches
- ✓ Pattern alteration and grading basics

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
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- Not connecting the topic to textile production or customer requirement.

Module 3

Cutting and sewing operations

Core explanation

- ✓ Fabric spreading, marker planning, cutting and bundling
- ✓ Machine selection, stitch class, seam class and needle selection
- ✓ Sewing defects: puckering, skipped stitch, uneven seam and corrective action
- ✓ Construction of sample garments and components

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Finishing, inspection and costing

Core explanation

- ✓ Trimming, pressing, folding, labelling and packing
- ✓ Measurement checking and visual inspection
- ✓ Defect classification and rework procedure
- ✓ Garment costing and practical record preparation

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

□□□□□□□: Practical answer □□□□□□□□□□□ definition □□□□□□□ □□□□. Process → parameter → defect → correction → record □□□□ flow □□□□□□□□□□.

Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Garment Cost = Fabric + Trims + Labour + Overhead + Profit

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Seam Allowance = Required finished size allowance for joining

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

Marker Efficiency % = Pattern Area / Fabric Area x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

SPI = Stitches per Inch

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Take measurements and prepare size chart
- ✓ Draft and cut a basic pattern
- ✓ Stitch selected garment component
- ✓ Prepare final inspection report

Important keywords

Pattern **Marker** **Grain line** **Seam** **SPI** **Pressing** **Inspection** **Costing**

Short questions

Q1.1	Explain: Elements of design: line, shape, colour, texture and balance	Answer with definition, process, application and one example.
Q1.2	Explain: Fashion figure, garment sketching and design board	Answer with definition, process, application and one example.
Q1.3	Explain: Fabric selection based on drape, end use and style	Answer with definition, process, application and one example.
Q2.1	Explain: Body measurement points and allowance	Answer with definition, process, application and one example.
Q2.2	Explain: Basic bodice, sleeve, skirt and trouser block principles	Answer with definition, process, application and one example.
Q2.3	Explain: Pattern layout, grain line, seam allowance and notches	Answer with definition, process, application and one example.
Q3.1	Explain: Fabric spreading, marker planning, cutting and bundling	Answer with definition, process, application and one example.
Q3.2	Explain: Machine selection, stitch class, seam class and needle selection	Answer with definition, process, application and one example.

Q3.3	Explain: Sewing defects: puckering, skipped stitch, uneven seam and corrective action	Answer with definition, process, application and one example.
Q4.1	Explain: Trimming, pressing, folding, labelling and packing	Answer with definition, process, application and one example.
Q4.2	Explain: Measurement checking and visual inspection	Answer with definition, process, application and one example.
Q4.3	Explain: Defect classification and rework procedure	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to Pattern. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Fashion Design & Apparel Manufacture Lab.
2. List four important keywords: Pattern, Marker, Grain line, Seam.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Fashion design preparation.
7. Compare two important concepts from Measurement and pattern drafting.
8. Explain the practical importance of Cutting and sewing operations in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Fashion Design & Apparel Manufacture Lab in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Fashion Design & Apparel Manufacture Lab.
2. Use keywords: Pattern, Marker, Grain line, Seam, SPI, Pressing, Inspection, Costing.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Fashion design preparation

- Elements of design: line, shape, colour, texture and balance
- Fashion figure, garment sketching and design board
- Fabric selection based on drape, end use and style
- Mood board, colour palette and design specification sheet

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Measurement and pattern drafting

- Body measurement points and allowance
- Basic bodice, sleeve, skirt and trouser block principles
- Pattern layout, grain line, seam allowance and notches
- Pattern alteration and grading basics

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Cutting and sewing operations

- Fabric spreading, marker planning, cutting and bundling
- Machine selection, stitch class, seam class and needle selection
- Sewing defects: puckering, skipped stitch, uneven seam and corrective action
- Construction of sample garments and components

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Finishing, inspection and costing

- Trimming, pressing, folding, labelling and packing
- Measurement checking and visual inspection
- Defect classification and rework procedure
- Garment costing and practical record preparation

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.